

Vibhawari Ramakrishna Wusirika

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Summary

Motivated Bioengineering graduate student with a background in Biotechnology, seeking to apply skills in medical device design and development. Resourceful and wanting to gain experience in CAD and collaborative critical thinking, aiming to translate scientific knowledge into innovative healthcare solutions.

Education

University of California, San Diego

Sept 2025-Expected Dec 2026

MEng Bioengineering with spec. in MDE (medical device engineering) | GPA: 4.0

Vellore Institute of Technology (VIT), Vellore, India

Sept 2021-May 2025

B.Tech Biotechnology | cGPA: 9.4/10

Projects

SAW-Actuated Microfluidic Gradient Module for Biohybrid Neural Interfaces, UCSD

Jan 2026-Present

- Created and resin printed microfluidic chip using Fusion 360
- Cultured and differentiated iPSCs to ONPs, developed ONP spheroids using EZSPHERE, demonstrated ONP patterning using SAW waves and neurite projection from ONPs towards BDNF gradient.
- Tested PODS-GFP diffusion gradient establishment by fluorescent microscopy and image analysis using Matlab.

Dissertation Student, Center for DNA Fingerprinting and Diagnostics (CDFD), Hyderabad

Jan-May 2025

Under Dr. Ashwin Dalal, Laboratory of Diagnostics

- Solved cases of thyroid dyshormonogenesis in 3 families by identifying missense, nonsense, and frameshift variants in the thyroglobulin gene by whole exome sequencing and verified by sanger sequencing.
- Designed primers for sanger sequencing and analyzed effect of variations on protein structure using DynaMut2.

Research Intern, Central University of Punjab, India

Jun 2024

Under supervision of Dr. Monisha Dhiman

- Cultured SHSY5Y cell line and evaluated the response of AGS cells to oxidative stress by NBT assay.
- Detected proteins in cell lysates by Western blotting analysis and performed PCR to detect the expression of Bax (pro-apoptotic) gene in cancer cells.

Summer Intern, Aragen Life Sciences

Aug-Sept 2023

Department of Drug Metabolism and Pharmacokinetics

- Performed in vitro stability assays including Plasma Protein Binding, Plasma Stability, Hepatocyte Stability, Microsomal Stability, and Kinetic Solubility Assays for novel chemical entities
 - Conducted LCMS and HPLC on novel chemical entities, adhering to standard operating procedures.
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Skills

Programming skills: Matlab programming

Design skills: Basic Fusion 360, Resin printing, Injection molding, Soldering

Software skills: Primer3, PrimerBLAST, PCR Primer Stats, DynaMut2, Spreadsheets

Wet lab skills: Hydrogel design, Rheometry testing, Cell culture, DNA and RNA isolation, Bacterial Transformation, Agarose Gel Electrophoresis, SDS PAGE, Western Blot, Protein isolation, ELISA, Clinical trial design

Leadership and Activities

Assistant Area Director of Finance and Member, Toastmasters VIT

July 2023-June 2024

- Oversaw the financial health of 10 clubs by training treasurers, and managed club bank account
- Enhanced public speaking skills through Toastmasters training, delivered ten speeches, mentored clubmates, organized events with area team, organized and participated in weekly club sessions

Volunteer, Youth Red Cross VIT

March 2022-June 2023

- Collected case details from patients requiring blood and assisted donors at annual Blood donation camps

Achievements in Ideathons

- Created and lead a team of 5 undergraduates to ideate two bio-inspired systems to win 2nd and 3rd place at Bio-Inspired Design (BID) Fest VIT 2022 and 2023
- Designed a sustainable alternative to conventional glitter with a team of 2 undergraduates, that was awarded first place in Ecolution ideation, VIT 2022